北京交通大學 BEIJING JIAOTONG UNIVERSITY

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Signal decomposition

- Into direct part and alternating part
- Into even part and odd part
- Into real part and imaginary part
- Into linear combinations of $\delta(t)/\delta[k]$

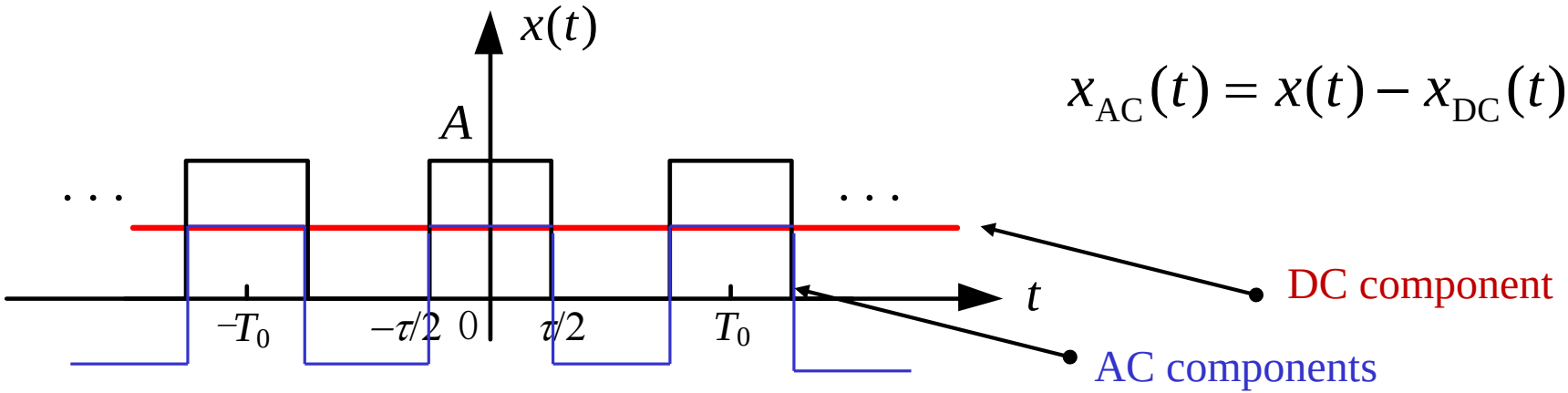
Into direct part and alternating part

C-T signal

$$x(t) = x_{\mathrm{DC}}(t) + x_{\mathrm{AC}}(t)$$

$$x_{\mathrm{DC}}(t) = \frac{1}{b-a}\int_a^b x(t)\mathrm{d}t$$

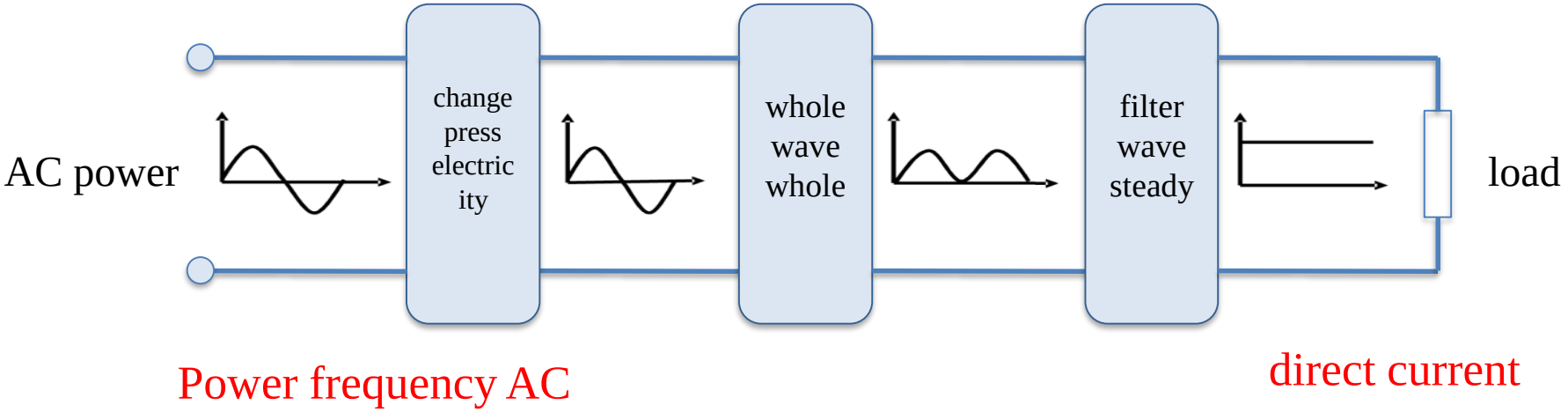
$$x_{\mathrm{AC}}(t) = x(t) - x_{\mathrm{DC}}(t)$$



$$x_{\mathrm{DC}}(t) = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)\mathrm{d}t = \frac{1}{T_0}\int_{-\tau/2}^{\tau/2} A\cdot\mathrm{d}t = \frac{A\tau}{T_0}$$

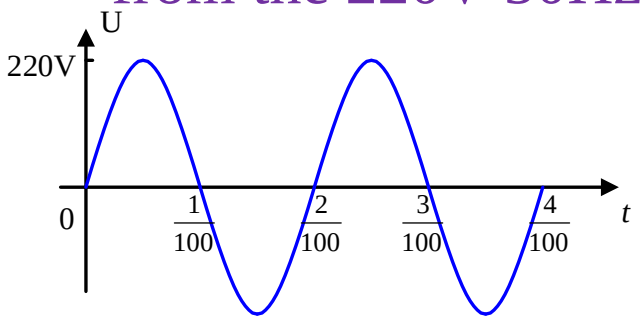
Into direct part and alternating part

A rectifier circuit is used to extract the required amplitude of the DC voltage from the AC voltage

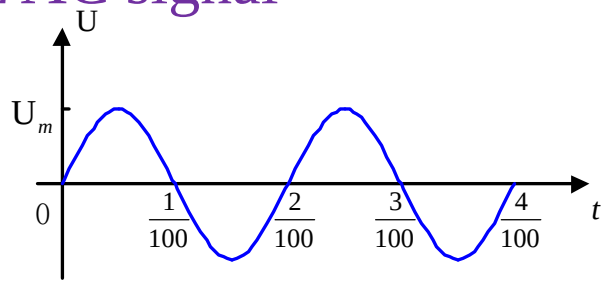


Into direct part and alternating part

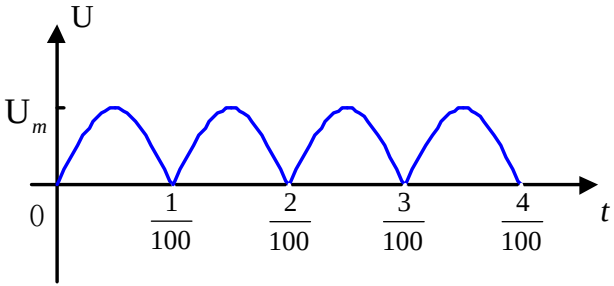
A full-wave rectifier circuit is used to extract the DC component from the 220V-50Hz AC signal



220V-50Hz AC signals



U_m-50Hz AC signals



U_m-50Hz
Full-wave rectifier signal

$$x_{\mathrm{DC}}(t) = \frac{1}{b-a}\int_a^b x(t)\mathrm{d}t \Rightarrow U_{DC} = 100\int_0^{\frac{1}{100}} U_m \sin(2\pi \times 50t)\,\mathrm{d}t = \frac{2}{\pi}\cdot U_m \approx 0.64U_m$$

Into direct part and alternating part

D-T signals

$$x[k] = x_{\mathrm{DC}}[k] + x_{\mathrm{AC}}[k]$$

$$x_{\mathrm{DC}}[k] = \frac{1}{N_2 - N_1 + 1} \sum_{k=N_1}^{N_2} x[k]$$

$$x_{\mathrm{AC}}[k] = x[k] - x_{\mathrm{DC}}[k]$$

Into even part and odd part

C-T signal

$$x(t) = x_e(t) + x_o(t)$$

Even Odd

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

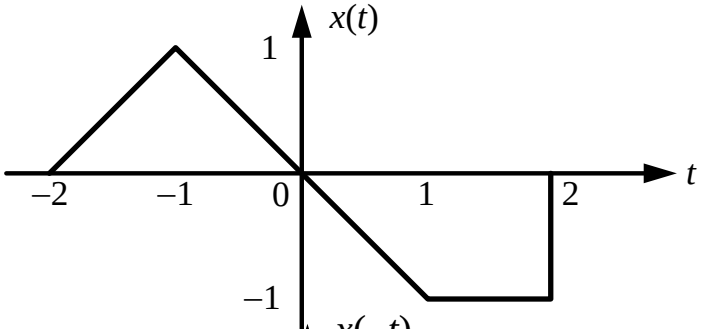
$$x_e(t) = x_e(-t)$$

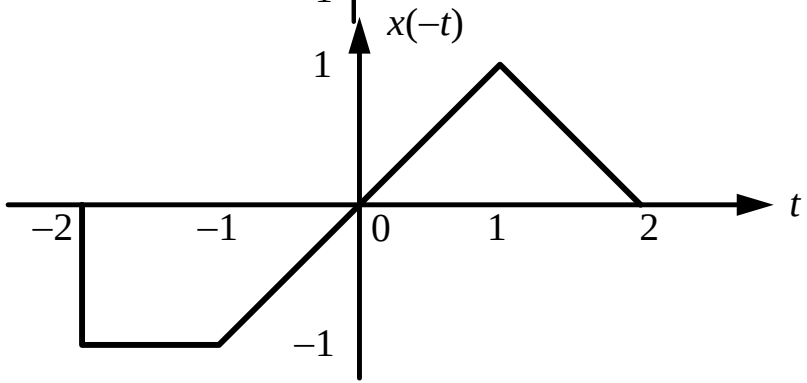
$$x_o(t) = -x_o(-t)$$

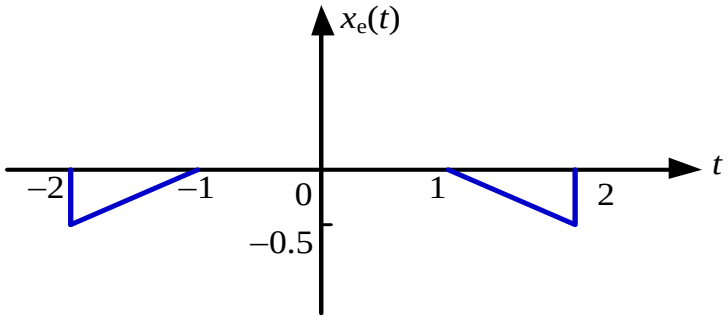
[Example] Plot the odd and even components of a C-T signal x(t).

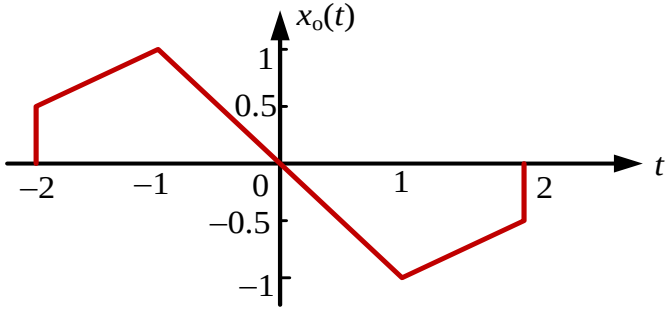
$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$









Into even part and odd part

D-T signal

$$x[k] = x_{\mathrm{e}}[k] + x_{\mathrm{o}}[k]$$

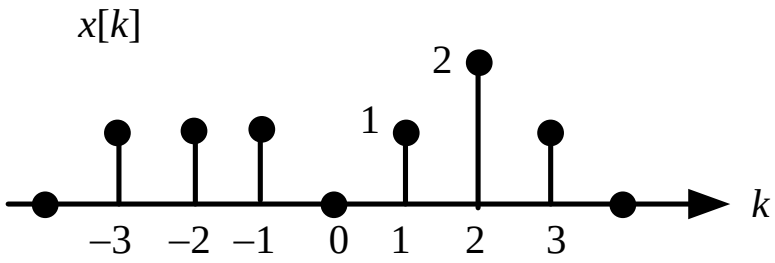
Even Odd

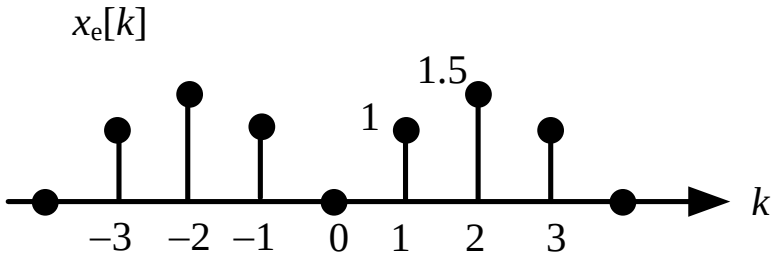
$$x_{\mathrm{e}}[k] = \frac{1}{2}\{x[k] + x[-k]\} \qquad x_{\mathrm{o}}[k] = \frac{1}{2}\{x[k] - x[-k]\}$$

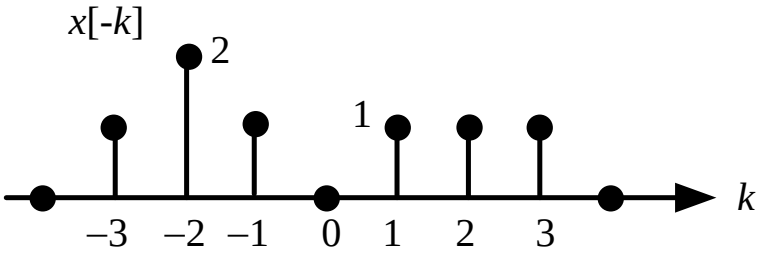
$$x_{\mathrm{e}}[k] = x_{\mathrm{e}}[-k] \qquad x_{\mathrm{o}}[k] = -x_{\mathrm{o}}[-k]$$

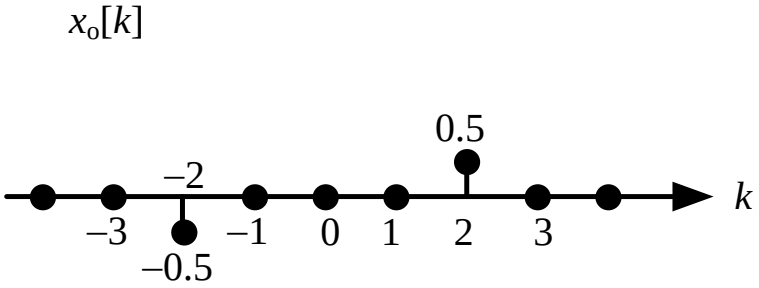
[Example] Plot the odd and even components of a D-T signal x(t).

$$x_e[k] = \frac{1}{2}\{x[k] + x[-k]\} \qquad x_o[k] = \frac{1}{2}\{x[k] - x[-k]\}$$









Into even part and odd part

The face has strong symmetry, but it is not completely symmetrical, and its parity component characteristics are prominent

Original image

Odd components

Even components

Into real part and imaginary part

C-T signal $$x(t) = x_{\mathrm{r}}(t) + \mathrm{j} \cdot x_{\mathrm{i}}(t)$$

Real part

imaginary part

$$x_{\mathrm{r}}(t) = \frac{1}{2}[x(t) + x^*(t)]$$

$$x_{\mathrm{i}}(t) = \frac{1}{2\mathrm{j}}[x(t) - x^*(t)]$$

$$x^*(t) = x_{\mathrm{r}}(t) - \mathrm{j} \cdot x_{\mathrm{i}}(t)$$

Into real part and imaginary part

D-T signal

$$x[k] = x_{\mathrm{r}}[k] + \mathrm{j} \cdot x_{\mathrm{i}}[k]$$

Real part

imaginary part

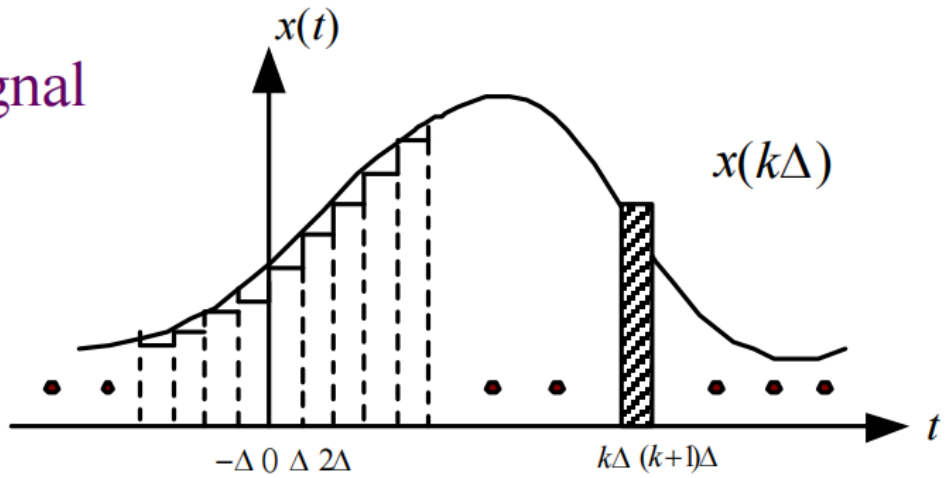
$$x_{\mathrm{r}}[k] = \frac{1}{2}\{x[k] + x^*[k]\}$$

$$x_{\mathrm{i}}[k] = \frac{1}{2\mathrm{j}}\{x[k] - x^*[k]\}$$

$$x^*[k] = x_{\mathrm{r}}[k] - \mathrm{j} \cdot x_{\mathrm{i}}[k]$$

Into linear combinations of δ(t)/δ[k]

C-T signal



A continuous signal is expressed as the superposition of the impulse signal

$$x(t) \approx \cdots + x(0)[u(t) - u(t-\Delta)] + x(\Delta)[u(t-\Delta) - u(t-2\Delta)] + \cdots$$

$$+ x(k\Delta)[u(t-k\Delta) - u(t-k\Delta-\Delta)] + \cdots$$

Into linear combinations of δ(t)/δ[k]

$$x(t) \approx \cdots + x(0)[u(t) - u(t-\Delta)] + x(\Delta)[u(t-\Delta) - u(t-2\Delta)] + \cdots$$

$$+ x(k\Delta)[u(t-k\Delta) - u(t-k\Delta-\Delta)] + \cdots$$

$$x(t) \approx \sum_{k=-\infty}^{\infty} x(k\Delta) \frac{[u(t-k\Delta) - u(t-k\Delta-\Delta)]}{\Delta} \Delta$$

$$x(t) = \lim_{\Delta \to 0} \sum_{k=-\infty}^{\infty} x(k\Delta)\delta(t-k\Delta)\Delta$$

$$= \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\mathrm{d}\tau$$

Into linear combinations of δ(t)/δ[k]

$$x(t)=\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\mathrm{d}\tau$$

Physical significance

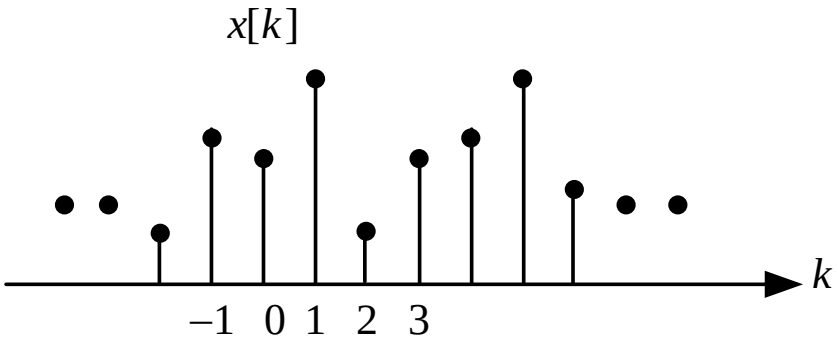
Different continuous signals can be expressed as a linear combination of the impulse signal and its time-shift, and different signals simply have different coefficients

Practical application

When solving the response of signal through LTI system, only the response of d (t) through the system needs to be solved, and then the response of signal x (t) acting on the system can be obtained by using the characteristics of LTI system.

Into linear combinations of δ(t)/δ[k]

D-T signal



$$x[k] = \mathrm{L} + x[-1]\delta[k+1] + x[0]\delta[k] + x[1]\delta[k-1] + \mathrm{L} + x[n]\delta[k-n] + \mathrm{L}$$

$$x[k] = \sum_{n=-\infty}^{\infty} x[n]\delta[k-n]$$

Into linear combinations of δ(t)/δ[k]

$$x[k]=\sum_{n=-\infty}^{\infty} x[n]\delta[k-n]$$

Physical significance

Different discrete signals can be expressed as a linear combination of the pulse train and its displacement, and different signals simply have different coefficients.

Practical application

When solving the response generated by the discrete signal through the LTI system, only the response generated by the solution pulse train through the system is required; Using the characteristics of the LTI system, the response generated by the signal x[k] acting on the system can be found

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!